Highlights

Automatically Resolving Conflicts Between Expert Systems: An Experimental Approach Using Large Language Models and Fuzzy Cognitive Maps

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- Adding logical ideas into a well-constructed expert system easily leads to dissonance
- Dissonance in Fuzzy Cognitive Maps was previously discussed but never operationalized
- We resolve dissonance in FCMs (a type of expert systems) using Large-Language Models
- Three case studies show that solving dissonance requires well-crafted causal edges
- Our work is provided open-source to support reuse for other cases or different LLMs

Automatically Resolving Conflicts Between Expert Systems: An Experimental Approach Using Large Language Models and Fuzzy Cognitive Maps

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Abstract

A mental model is an individual's internal representation of knowledge that enables reasoning in a given domain. Cognitive dissonance arises in a mental model when there is internal conflict, causing discomfort, which individuals seek to minimize by resolving the dissonance. Modelers frequently use fuzzy cognitive maps (FCMs) to represent mental models and perspectives on a system and facilitate reasoning. Dissonance may arise in FCMs when two individuals with conflicting mental models interact (e.g., in a hybrid agent-based model with FCMs representing individuals' mental models). We define cognitive dissonance for FCMs and develop an algorithm to automatically resolve it by leveraging large language models (LLMs). We apply our algorithm to three real-world case studies and find our approach can successfully resolve the dissonance, suggesting LLMs can broadly resolve conflict within expert systems. Additionally, our method may identify opportunities for knowledge editing of LLMs when the dissonance cannot be satisfactorily resolved through our algorithm.

Keywords: Cognitive Dissonance, Fuzzy Cognitive Maps, Generative AI, Large Language Models, Mental Model, Natural Language Generation

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1. Introduction

Cognitive dissonance is a state of psychological discomfort resulting from incompatibilities in an individual's *mental model* [1], the mental framework used by individuals to facilitate reasoning and understanding. The theory of cognitive dissonance proposes that beliefs, behaviors, or desires (often represented as the concept nodes of an individual's mental model) can either agree or conflict with each other. These conflicts cause dissonance, which are uncomfortable; hence, individuals seek to minimize dissonance [1]. For example, an individual may publicly lie about their opinion on an issue, creating a dissonance between their privately held and publicly stated beliefs with a larger difference in beliefs causing a greater dissonance and more intense pressure to resolve it [1, 2].

Fuzzy Cognitive Maps (FCMs) are weighted, directed graphs that represent degrees of causality between different objects or concepts. FCMs extend Cognitive Maps by introducing fuzziness when the classes and causes of a knowledge domain are uncertain [3]. They are used as a computational representation of mental models as their use of fuzzy logic enables handling uncertainty and imprecision in the relationships between concepts [4]. In particular, FCMs and their extensions are widely used as expert systems [5, 6, 7, 8]. Two FCMs can interact, thus representing interactions between two expert systems (e.g., as part of an ensemble of FCMs [9]) or between the cognitive models of two agents [10, 11, 12]. As a result of an interaction, new concepts and relationships can be introduced to each FCM. If these changes alter the output of the FCM, then the conclusions that an individual previously held about the world have been drastically altered, hence creating a cognitive dissonance.

Although the notion of cognitive dissonance has been discussed for FCMs [13], it has never been operationalized. Frameworks that allow for the interaction of FCMs have been limited to changing existing weights, instead of altering the structure of a model [14]. This is a critical research gap since interactions between expert systems can have wider implications than only changing weights, and these implications may leave one system in an internal state of dissonance without methods to address it. The *main contribution of this paper* is enabling FCMs that represent mental models to resolve the dissonance created through interaction by adding new concepts or relationships. Our work investigates multiple agents and their knowledge represented as expert systems, examines where dissonance occurs when agents interact, and

automatically resolves the dissonance. For example, two agents with mental models on obesity may interact, creating dissonance among the agents (Figure 1). In this figure, Agent A thinks exercise increases physical health, which decreases obesity, whereas Agent B believes that exercise increases musculoskeletal disorders, increasing obesity. After their interaction, Agent B learns that exercise will lower obesity and will not increase musculoskeletal disorders. To resolve the dissonance introduced by the added edge from exercise to obesity in the mental model of Agent B, the pathway from obesity to musculoskeletal disorders to pain from exercise will gradually disappear.

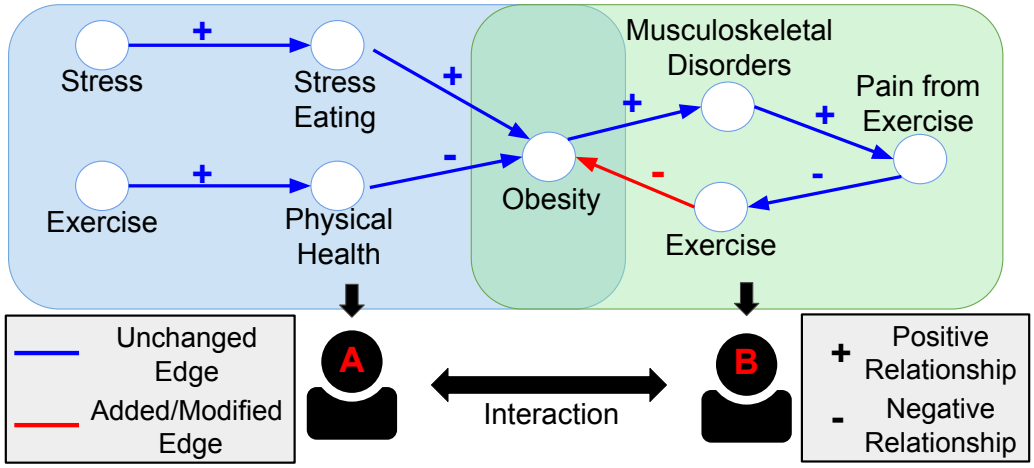

Figure 1: Example of dissonance in mental models.

Our work utilizes the ability of Large Language Models (LLMs) to solve complex tasks, given their training on vast amounts of diverse data. LLMs are broadly used for numerous downstream tasks (e.g., summarization [15, 16, 17]), including to explain FCMs and related network-based mental models [18, 19] or to derive such models from text [20]. Our work is the first to use LLMs to solve the dissonance introduced by the interaction between two expert systems. Our work answers three research questions:

(Q1) Can a LLM modify an expert system to reduce the dissonance introduced by another expert system?

(Q2) What is an optimal LLM configuration to reduce dissonance?

(Q3) In the process of reducing dissonance, how is an expert system modified?

The remainder of the paper is organized as follows. In section 2, we introduce foundational concepts of FCMs, LLMs, and LLMs for editing knowledge bases. We define our methods for quantifying cognitive dissonance, resolving it, and applying these methods to three real-world case studies in section 3. In section 4, we present our results from these three case studies. Finally, in section 5, we discuss the importance of our results and provide directions for future work.

2. Background

2.1. Fuzzy Cognitive Maps

Fuzzy Cognitive Maps (FCMs) are interpretable discrete simulation models that simplify a system of interest by focusing on the core concepts and their causal relationships [3]. They can be built through semi-structured interviews with subject matter experts [21, 22] or machine learning-based approaches with historical data on the system [23, 24, 25]. FCMs represent the system as a directed, weighted graph, where each concept is clearly labeled, enabling interpretation of the edges. Each concept is assigned a value from [0, 1], where 0 indicates its complete absence (e.g., no obesity) and 1 indicates its complete presence (e.g., as much obesity as possible). Each edge is assigned a weight from [-1, 1], where an edge weight of -1 indicates a strongly negative causal relationship (e.g., as physical activity increases, diabetes decreases), 0 specifies no causal relationship, and 1 implies a strong positive causal relationship (e.g., as body weight increases, diabetes increases).

FCMs perform simulation in discrete iterations, and concept values are updated synchronously. Simulation continues until a set of output concepts (e.g., obesity) changes by less than a user-defined threshold or the maximum number of iterations is reached. Mathematically, an FCM at an iteration t can be represented as the tuple (C^t, W, f), where C^t is a vector storing the concept values at the t-th iteration, W is the adjacency matrix specifying the edge weights between concepts, and f is a threshold function [4]. Common choices of f are the sigmoid function $f(x) = \frac{1}{1+e^{-sx}}$, where s controls the steepness of the function, and the hyperbolic tangent function $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, which limit the concept values to [0, 1] and [-1, 1], respectively [26, 27]. There are two popular update equations for FCMs [26, 27]. First, there is the FCM 'without memory' that gives a value for a concept C_j at an iteration $t+1$ as

$$C_j^{t+1} = f\underbrace{\left(\sum_{\substack{i=1\\ i\neq j}}^{n} W_{ij} \times C_i^t\right)}_{\text{total input to node}}, \tag{1}$$

where W_{ij} is the weight of the edge from the concept C_i to C_j and C_i^t is the value of the concept C_i at iteration t. The condition $i \neq j$ prevents the previous value of the concept from affecting its current value. Thus, an alternate update equation 'with memory' was proposed to allow the previous value of the concept to affect the present value [28]:

$$C_j^{t+1} = f\left(\underbrace{\left(\sum_{\substack{i=1\\ i\neq j}}^{n} W_{ij} \times C_i^t\right)}_{\text{total input to node}} + \underbrace{C_j^t}_{\text{memory}}\right) \tag{2}$$

2.2. Large Language Models

Thanks to technical advances such as the 2014 attention mechanisms and its expansion into a deep learning architecture (known as transformer) in 2017, Natural Language Generation (NLG) shifted from a historical reliance on hand-crafted rules and modules, embracing a new era of fast-paced development focused on training models from data [29]. Although OpenAI's GPT is the poster child of these new developments, there is a large variety of LLMs such as LLaMa (from Meta), Gemini (from Google DeepMind), or Claude (from the startup Anthropic). Data plays a central role in LLMs, not only to train the models but also to evaluate their performance or refine their understand of specific application domains (i.e., fine-tuning) [30]. As models grew to billion or even trillon parameters, fine-tuning a large number of parameters resulted in spiraling computation and storage costs. New approaches thus include the optimization of a few parameters [31] as well as an examination of whether a few well-crafted samples would provide sufficient background (i.e., few shot learning) [19].

To successfully integrate an LLM in an application, users need to carefully design a *prompt* [32]. Users write a prompt to express a specific task, such as retrieving information, improving, creating, or summarizing [33]. The

prompt can also specify the approach to text generation, for instance by breaking down a complex problem into steps (i.e., chain-of-thought reasoning [34]). It can be challenging to engineer the right prompt and notice issues in the generated output. For example, explanations delivered through chain-of-thought reasoning may appear convincing, while misrepresenting the actual reason for which the model reached a decision [35]. Furthermore, recent works have suggested that an LLM does not inherently represent characteristics of a desired output (e.g., 'write motivational text'); rather, it transfers styles (e.g., 'write like Oprah') [36]. Assigning a *role* is another powerful way to shape the output of an LLM, as a study showed that asking an LLM to behave as an expert resulted in better performances than when 'impersonating' a non-expert [37]. The several parts that compose a prompt (e.g., examples, counter-examples) may interact in shaping the output's quality, and these interactions may have the counter-intuitive effect of decreasing performances when using longer prompts [38]. In addition, the output is also shaped by model-specific parameters, which can control either creativity (e.g., the *temperature* setting in GPT or Gemini) or the expected scope (e.g., setting a maximum number of tokens in the generated output).

2.3. Large Language Models for Completion of Smaller Models

An FCM rarely exceeds a hundred nodes, and FCMs obtained from participatory modeling tend to be contain a few dozen nodes. In this paper, we use LLMs as a larger *knowledge repository* to supplement the smaller FCMs (and solve their conflicts). This is a *Knowledge Base Completion* (KBC) task, for which LLMs have been employed in several studies. We thus provide a succinct background to relate our use of LLMs to the broader literature on KBC, with sample studies characterized in more details through Table 1. In several studies, the rationale for KBC is to reduce the resources and time needed to construct a knowledge base, especially compared to manual construction. For example, GPT-3 added 27,000,000 statements with 90% precision to Wikidata, the largest public knowledge base, outperforming smaller language models such as BERT while costing an average of $0.007/statement in contrast to $2.00/statement for manual curation [39]. Additionally, language models such as Prix-LM have constructed and enriched multilingual knowledge bases, outperforming state-of-the-art knowledge graph embedding models such as RotatE on link prediction and mBERT on cross-lingual identity linking [40].

Table 1: Related works on LLMs for the completion of smaller models.

Refs	LLM	Usage	Application	Fine-tuning Strategies	Evaluation Metrics
[39]	GPT-3	KBC	Wikidata KB		Precision, Recall, F1 Score
[40]	Prix-LM	KBC	KB Construction Tasks	Training on knowledge triples	Link Prediction Scores (Hits@1, Hits@3, Hits@10), LR-XEL benchmark, XL-BEL, MRR
[41]	KG-BERT, GPT-2	KGC	Triple Classification sub-task on datasets WN11 and FB13	Training on knowledge triples	Accuracy, Precision, Recall, F1 Score
[42]	LLaMA, ChatGLM	KGC	Triple Classification, Relation Prediction, Entity Prediction	ChatGLM-6B fine-tuned with P-tuning v2, LLaMA fine-tuned with LoRA	Accuracy (Hits@1)
[43]	KoPA	KGC	Triple Classification	Training on seen knowledge triples	Accuracy, Precision, Recall, F1 Score

The strand of literature closer to our study is the application of LLMs for Knowledge *Graph* Completion (KGC). This task involves completing the links, relations, and facts of a Knowledge Graph. FCMs share a few characteristics with Knowledge Graphs: edges represent directed relationships between the real-world entities stored as labeled nodes [41]. However, it is important to stress that *Knowledge Graphs differ from FCMs* for several reasons. An FCM is a simulation model, so nodes and edges have quantita-

tive values. In contrast, labels on the edges of knowledge graphs specify the type of relationship [41]. Leveraging LLMs to complete knowledge graphs can efficiently resolve missing or inconsistent graphs [41, 42]. One challenge identified in maximizing the performance of applying LLMs in this context was ensuring the applied LLMs were aware of the structure of the graph [43]. For a recent review, we refer the reader to section 5 in [44].

2.4. Knowledge Editing Large Language Models

While the general knowledge or text processing ability of LLMs can allow them to inject new content in knowledge bases, the *explicit* facts stored in a knowledge base can also improve LLMs and lead to more informed responses [45]. While improving an LLM is not the purpose of this article, our algorithm can contribute to identifying domains where improvements are needed; hence, we briefly contextualize this research area. *Knowledge Editing* (KE) is the process of modifying and updating the internal knowledge of an LLM, allowing the LLM to stay accurate and reliable. Some of the interest in this field has been motivated by viewing KE as a less resource-intensive approach than retraining to deal with false or outdated knowledge. In particular, the notion of *lifelong model editing* posits that edits will be continuously needed throughout a model's deployment [46]. Significant performance losses are a potential concern (e.g., a gain in memorizing specific facts at the expense of generalization performance), as there are rippling effects when editing a complex object such as a LLM [47, 48]. While editing methods such as ROME can identify the neurons involved in specific predictions [49], emerging research suggests that *locating* the parts of a model that produce a behavior and *editing* to modify the behavior are more disjoint tasks than expected. For example, an output may seem primarily attributable to a specific layer of the network (locating), but the output may be best modified by making changes in a different layer (editing) [50]. Regardless of the methods involved in fixing the problem, an important starting point is to identify a need for fixes. A variety of prompt-based bias datasets have shown that LLMs needed to be fixed in terms of geography as well as "race, ethnicity, socioeconomic status, culture, and politics" [51].

3. Methods

We quantify cognitive dissonance (section 3.1) then propose a method to resolve it leveraging LLMs (section 3.2). In section 3.3, we explain our em-

pirical evaluation with three case studies using the FCMs from peer-reviewed articles and several LLM configurations.

3.1. Quantifying Cognitive Dissonance

Cognitive dissonance can be intuitively explained as follows. Prior to interacting with others, an individual held some view of the world: given a scenario, they would reach certain conclusions. For example, an individual could justify biking by stating that it leads to better physical health. Through social interactions, new concepts and relationships can enter the picture. An individual could be made aware that they live in a very polluted environment, and that exercising in this setting will lead to inhaling more particles, which is detrimental to their health. This causes a cognitive dissonance: the individual engaged in an activity that they liked and found ways to justify it, but the new information would force them to make a different conclusion. This story shows four key components: an initial worldview, an initial conclusion, a modified worldview, and a potentially altered conclusion.

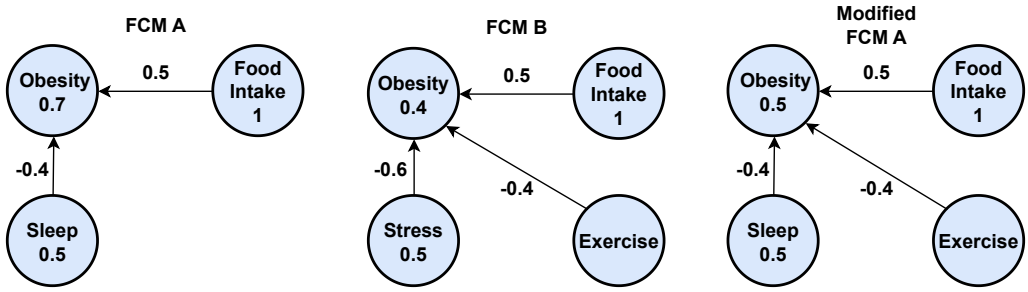

Figure 2: Example dissonance in FCMs.

Mathematically, we denote O_1 as the output of the original FCM (i.e., the value for a node of interest after iterations end) and O_2 as the output of the modified FCM. The cognitive dissonance between FCMs is measured as the distance between their outputs, $|O_1 - O_2|$. Note that the initial input vector is the same in both FCMs: we check if conclusions are changing with respect to the *same* scenario. For example, if an FCM models the amount of obesity and has 0.7 as the initial final value and 0.5 as the final value after changing, the cognitive dissonance for the FCM is $|0.7 - 0.5| = |0.2| = 0.2$ (Figure 2).

While there are many methods to compare FCMs, they usually do so on the basis of the *structure*, e.g. by comparing the number of concepts

or computing a Graph Edit Distance [52, 53]. By comparing simulation outcomes, we examine how the complete set of structural changes in an FCM could change the conclusion and hence create a discomfort that must be resolved.

3.2. Resolving Cognitive Dissonance

We propose a seven-step process to resolve the cognitive dissonance within an FCM by leveraging an LLM. Our process is depicted in Figure 3 and we fully support replicability effort by providing our documented code on a third-party permanent repository at `https://osf.io/z543j//`

(1) We send a prompt[1] to the LLM describing the inputs, defining cognitive dissonance, and instructing it to resolve the dissonance by adding new edges or nodes. We set the role and hence expectations as being an assistant who understands FCMs. The LLM can add *new* nodes (with initial values) and edges (with a weight). However, we explicitly prohibit the LLM from *changing* node values (i.e., changing the input scenario for the simulation) or edge weights (i.e., the strength and direction of the causal relationships) in the original and modified FCMs. As evidenced in Supplementary Material 1.1, this restriction is needed; otherwise, the LLM would distort the simulation scenario and existing causal relationships. Moreover, we instruct the LLM to only add edges that make sense for the FCM (e.g., where a positive causal relationship exists between the two concepts). Again, this instruction is necessary to avoid create a vast number of illogical edges with small weights (see Supplementary Material 1.2). We included examples in the prompt based on our prior work on LLMs and mental models, which showed that examples provided useful guidance [20].

> You are a helpful assistant who understands Fuzzy Cognitive Maps (FCMs). You will be given eight inputs in the following form:
> Adjacency Matrix: *adjacency_matrix*
> Concept Names: *concept_names*
> Initial Concept Values: *initial_concept_values*

[1]Depending on the implementation, the user input may be sent as a separate user prompt in addition to the system prompt, or it may be combined with the system prompt and sent as one collective prompt.

Original Result: *original_result*

New Concept Names: *new_concept_names*

Modified Adjacency Matrix: *modified_adjacency_matrix*

Modified Concept Values: *modified_concept_values*

Modified Result: *modified_result*

adjacency_matrix is an array depicting the adjacency matrix between the concepts in the FCM with each entry indicating the edge weight. *concept_names* is a list of the nodes/concepts in the FCM in the same order as the adjacency matrix. For example, the first concept in *concept_names* will be the first row and column in the *adjacency_matrix*. *initial_concept_values* is an array containing the initial values provided to the FCM. *original_result* is a floating point number specifying the result of performing simulation with the initial *adjacency_matrix* and will be between 0 and 1. *new_concept_names* is a list containing the names of new concepts added to the FCM. *modified_adjacency_matrix* is an array depicting the modified adjacency matrix. Edge weights may be modified and rows and columns may be added from the original *adjacency_matrix*. New columns and rows will correspond to the concepts in *new_concept_names*. *modified_concept_values* is an array containing the initial values for any new concepts added. For example, if one new concept is added (shown by an additional row and column in *modified_adjacency_matrix* and one value in *new_concept_names*), there will be one entry in the array. *modified_result* is a floating point number specifying the result of performing simulation with the modified adjacency matrix/FCM given the changes.

Your goal is to determine edges and nodes you can add to the FCM to minimize the effect of the changes, which is the absolute difference between *modified_result* and *original_result* (i.e., |*modified_result* - *original_result*|). Please make sure the new nodes, edges, and edge weights make sense for the FCM (e.g., a positive causal relationship exists between the two concepts). You cannot modify existing edge weights. Please return the changes you want to make to minimize the effect of the changes as a list of edges to add with their weights: [("source", "target", "weight": weight), ..., ("sourceN", "targetN", "weight": weightN)]. You may add new concepts to the FCM. If you do please return their initial values as a separate list after the list of new edges in the form: [("concept", initial_value), ..., ("conceptN", initial_valueN)]. Note, that

you cannot modify existing concept values in *initial_concept_values* or *modified_concept_values*. Return only the two lists; nothing else should be included in your response for brevity.

For example, return:

New Edges List: [("Oil Spill", "Pollution", "weight": 1), ("Climate Change", "Pollution", "weight": 0.6), ("Deforestation", "Pollution", "weight": 0.4),]

New Concept Values: [("Oil Spill", 0.7), ("Climate Change", 0.1), ("Deforestation", 0.1)]

No text should be included in your response other than the list of edges and their weights and new concept values

We parse the response from the LLM for (2) the returned list of new edges (specified by New Edges List in the response) and (3) new concepts' initial values (indicated by New Concept Values in the response). The list of new edges also specifies the new nodes, if any, since a node is created if it is an edge's source or target node and is not in the FCM. (4) We add the new nodes and edges to the FCM. (5) We perform a simulation using the FCM as modified by the LLM, to assess how much of the dissonance is resolved. For example, if the original dissonance value is 0.2 and the dissonance is 0.3 after implementing the changes, the LLM actually *increased* the dissonance in the model. On the other hand, if the resulting dissonance is 0.1, 50% of the dissonance was resolved. (6) If the changes resolve part of the dissonance, we store the changes, increment the number of successful attempts, and restart the algorithm. If they do not, we disregard the changes and increment the number of unsuccessful attempts. (7) We check if the number of successful and unsuccessful attempts to resolve the dissonance are below their respective user-defined thresholds to prevent the LLM from performing more replications than desired or failing to resolve the dissonance more often than would be tolerable. For example, for a given case study and LLM configuration, the LLM may not be able to resolve the dissonance, and a user may desire the algorithm to terminate after five failed attempts. Thus, if either threshold is exceeded, we return the current results. Otherwise, we go through the entire process again for another iteration. Since there is randomness in the answers produced by the LLM, asking a question repeatedly can produce a better outcome.

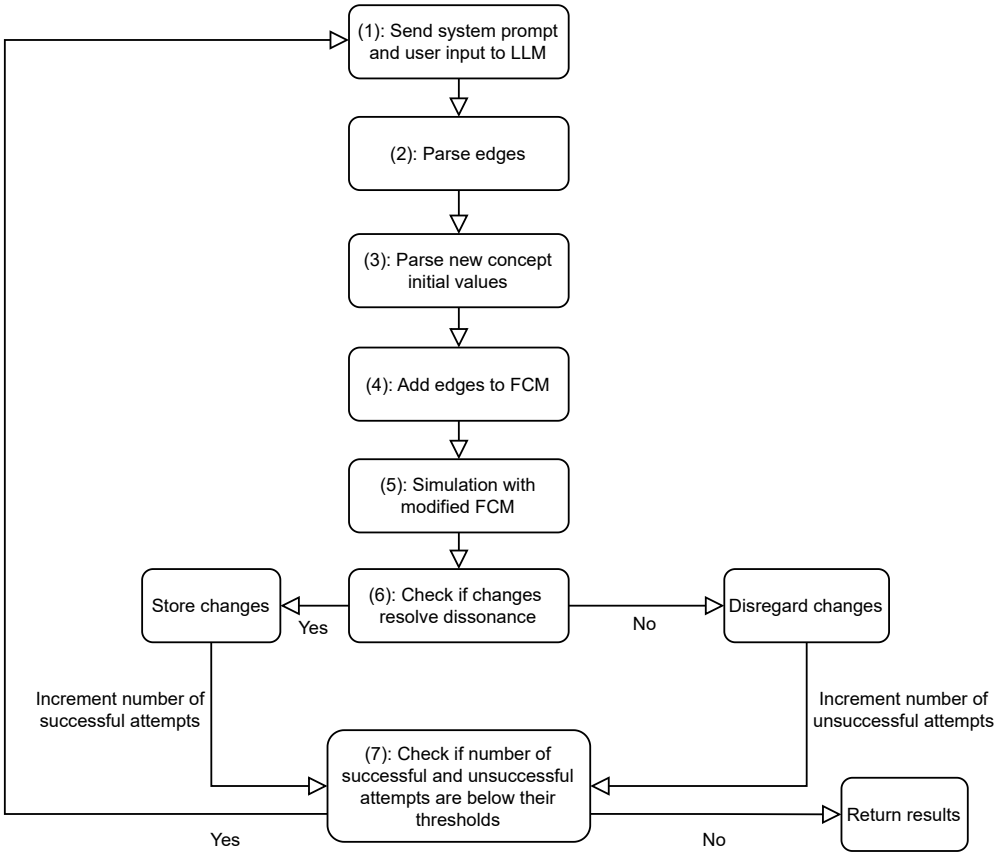

Figure 3: Algorithm to resolve cognitive dissonance.

3.3. Case Studies

We leverage GPT-4 and GPT-4 Turbo[2] with temperature values of 0, 0.25, 0.5, 0.75, or 1 (a higher temperature allows more variability in the response) via the OpenAI API in Python 3.10.12 to resolve the cognitive dissonance between an FCM for three real-world case studies (Table 2) and explore the impact of different LLMs and temperature values on the dissonance resolved. First, the FCM on obesity investigates how Canadian adults can collaboratively lower the prevalence of obesity, finding that real-world social structures reduce the effect of obesity reduction interventions com-

[2]Specifically, we use gpt-4-1106-preview. Experiments were performed in December, 2023.

pared to random connections among peers [54]. Second, the insurgency case study uses FCMs to simulate the social and geographical factors determining rebelliousness, supporting the analysis of military evaluations of potential counterinsurgency scenarios [55]. Third, the diabetes case study creates a FCM representing the knowledge of the traditional scientific community on the determinants of diabetes among Canadian aboriginals and an FCM encompassing the views of the Canadian aboriginal community on the same problem [56]. We use the former as the base case.

Table 2: Characteristics of the three case studies.

Case study	Number of nodes	Number of edges	Update equation	Threshold function	Reference
Obesity	13	20	'with memory'	Hyperbolic tangent	[54]
Insurgency	11	18	'with memory'	Hyperbolic tangent	[55]
Diabetes	18	29	'without memory'	Sigmoid	[56]

For each study, we measure the value of the output concept of interest (obesity, rebelliousness, and diabetes, respectively) in the original FCM and the modified FCM after introducing dissonance by adding new edges and nodes. Specifically, we added three edges to each case study, three nodes to the obesity and insurgency case studies, and two nodes to the diabetes FCM. For the obesity case study, we added sleep quality, healthcare access, and stress management as new concepts and edges from them to obesity with a weight of negative one, indicating they all decrease obesity. Issues in each of these three factors can increase the risk of obesity [57, 58, 59], but individuals are not necessarily aware that these factors play a role [60]. We thus simulate the realistic scenario in which proven contributors to obesity enter the mental model of an individual.

Second, for the insurgency FCM, we added the concepts of government legitimacy perception, government media and propaganda, and social cohesion with negative edges to rebelliousness with weights of -0.6, -0.4, and -0.4, respectively. The effects of the symbolic capital of government legitimacy onto insurgency dynamics have been well-studied [61], and government propaganda or social cohesion are also established topics in this field [62, 63]. It is thus plausible that one expert would bring these concepts to the attention of another. Finally, we added organ cleansing and healthy diet to the diabetes FCM, and we added edges from them and physical activity to diabetes with an edge weight of negative one (Figure 4). In this case, we took the

three concepts and their weighted relationships from the FCM representing the views of the Canadian aboriginal community. This case study thus represents the situation where an academic expert and a community member discuss their perspectives.

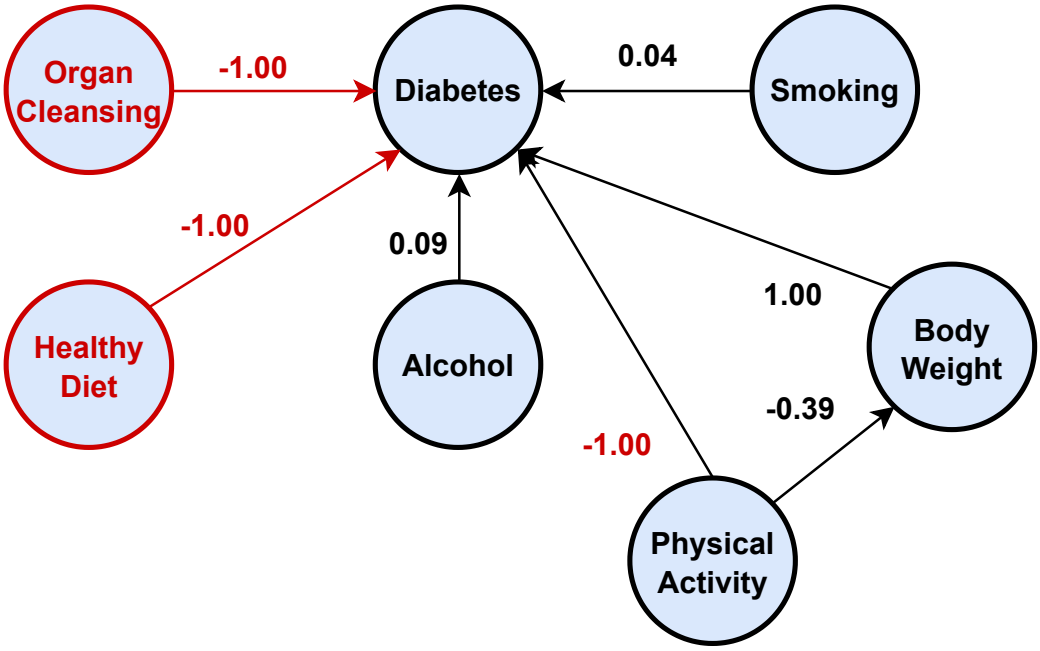

Figure 4: A subset of the traditional scientific community's diabetes FCM in black and the changes added from the Canadian aboriginal community's FCM shown in red.

We record the percentage of dissonance resolved, the number of new nodes and edges added, and the percentage of edges that appear logical for each combination of case study and model configuration. We define the percentage of dissonance resolved as $\frac{D_a}{D_o} \times 100$, where D_a is the dissonance after running the algorithm and D_o is the original dissonance introduced by the changes. To determine the percentage of logical edges, we count the number of logical and illogical edges (Figure 5) and divide the number of logical edges by the total number of edges suggested by GPT. We categorize an edge as logical if its directionality and type (+, -) are either *(i)* present in the original FCM (since the models were validated and subjected to peer-review evaluation) or *(ii)* approved by the lead author for the obesity and insurgency case studies and a public-health domain expert for the diabetes case study. The number of replications in each case study was determined using the Confidence Inter-

val method [64], which is a statistical determination of the number of runs in order to bound a measurement of interest (the percentage of dissonance resolved) within a desired interval. A tighter confidence interval requires more runs. In the case studies on nutrition and obesity, we used a 90% confidence interval. We used 80% for GPT-4 on the insurgency case study due to the high variability observed in the samples.

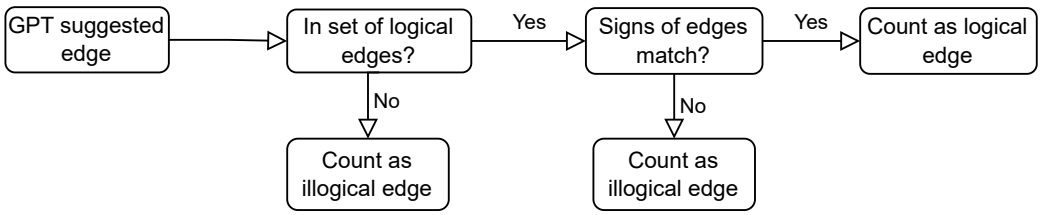

Figure 5: Process to count number of logical and illogical edges.

4. Results

Adding the three new nodes and edges to each case study introduced dissonance to each model (Table 3), demonstrating that *the addition of logical ideas into a well-constructed expert system can easily lead to dissonance*. Our first research question sought to reduce this dissonance via our proposed algorithm. Our results demonstrate the feasibility of the algorithm (Figure 6), as it solves at least 85% of the dissonance (for obesity) and up to 99% (for insurgency). As a follow-up, our second question examines which LLM configuration is optimal by using different models and different temperature settings. Our results show that the choice of a model matters, but the optimal choice depends on the application. For example, GPT-4 vastly outperformed GPT-4 Turbo across all temperature settings for the obesity case study (Figure 6 and Table A.4). However, the situation was different in the other two case studies, where we also observe the effect of temperature. By carefully choosing the temperature, GPT-4 Turbo achieved the best score (99%) on insurgency whereas an optimized GPT-4 only reached 76%. Interestingly, we note that neither GPT-4 nor GPT-4 Turbo could resolve any dissonance with a temperature setting of 0 in this case study, which indicates that some creativity was required.

Our third research question focused on the changes made to an expert system in order to address the dissonance. The main change was the addition

Table 3: The original results, results with dissonance, and the optimal average percent of dissonance resolved (i.e., the best result) for each of the three case studies.

Case study	Original result	Result with dissonance	Dissonance resolved
Obesity	0.908	0.505	85.500%
Insurgency	0.816	0.564	99.210%
Diabetes	0.693	0.287	91.670%

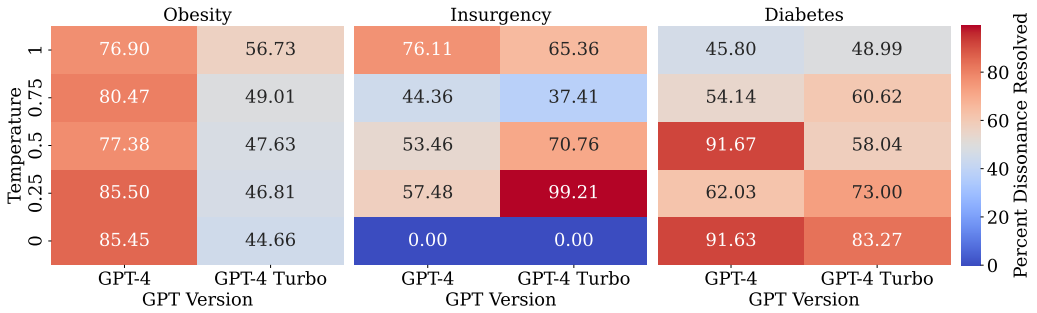

Figure 6: Mean percent dissonance resolved.

of causal edges. We note again that there is a joint effect from the model, temperature setting, and case study (Figure 7 and Table A.5). Across these settings, we observe that GPT-4 tends to add more edges than GPT-4 Turbo. While adding more edges is *often* associated with better performances in terms of lowering dissonance, we note that it is not always the case. For example, in the obesity case study, GPT-4 at temperature 0.75 adds the most edges (6.55 on average) but it achieves a lower performance (80%) than at temperature 0.25 when it adds fewer edges (4.64). This indicates that bridging dissonance is not merely a matter of 'saturating' a model with new edges, but rather a careful construction. Although our prompt allowed for the addition of concepts, GPT-4 and GPT-4 Turbo rarely added new nodes with the exception of GPT-4 Turbo with temperature 0.25 adding 0.5 new nodes on average for the insurgency case study (Figure 8 and Table A.5).

As part of our investigation into the structural changes made to an expert system, we checked whether the proposed new edges were logical, that is, justifiable in this context. Intuitively, we may expect that an increase in the temperature setting (leading to more 'creativity') would tend to spawn more illogical edges. However, this trend is only observed in one of six settings (GPT-4 Turbo in obesity). GPT-4 Turbo's edges were more logical than

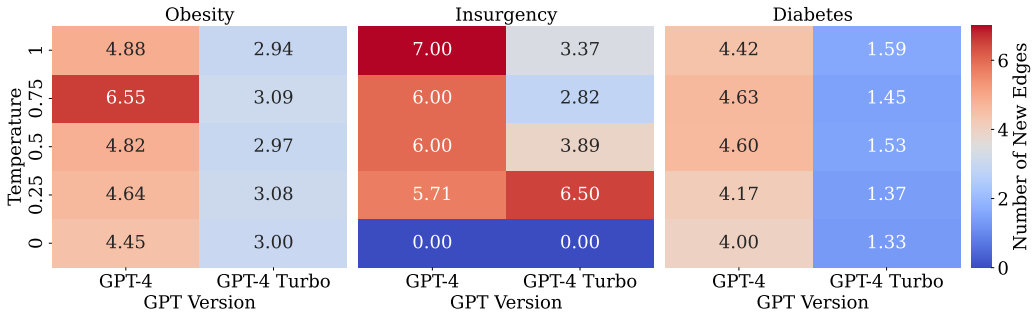

Figure 7: Mean number of new edges.

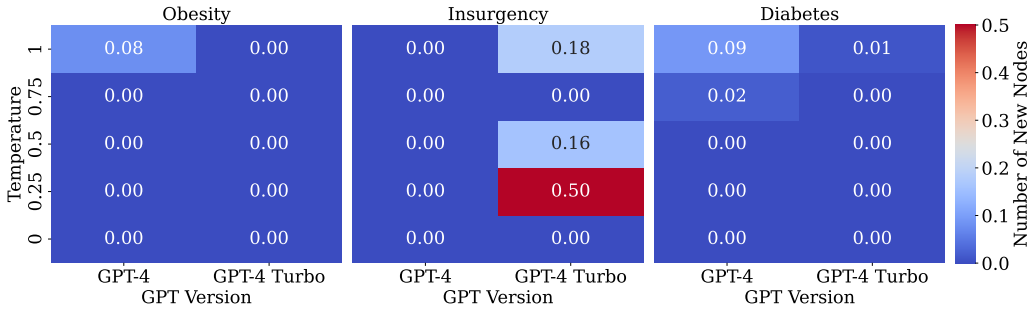

Figure 8: Mean number of new nodes.

GPT-4's for all case studies and temperatures other than for the diabetes case study with temperature 0 and 0.25 (Figure 9 and Table A.6). Nearly all of GPT-4 Turbo's changes were logical on the obesity case study. The insurgency and diabetes case studies were more challenging to resolve, which may be attributable to several reasons such as a lower familiarity of the models with these applications (e.g., their training set may include obesity more often than diabetes) or fewer potential edges being labeled as logical.

5. Discussion

Although the concept of cognitive dissonance has been discussed for FCMs, the operationalization of this process was unexplored. Our approach is the first to resolve dissonance and conflicting viewpoints in FCMs representing mental models by introducing new relationships and concepts. The successful application of our method to three case studies demonstrates that LLMs can resolve the cognitive dissonance between two FCMs or mental

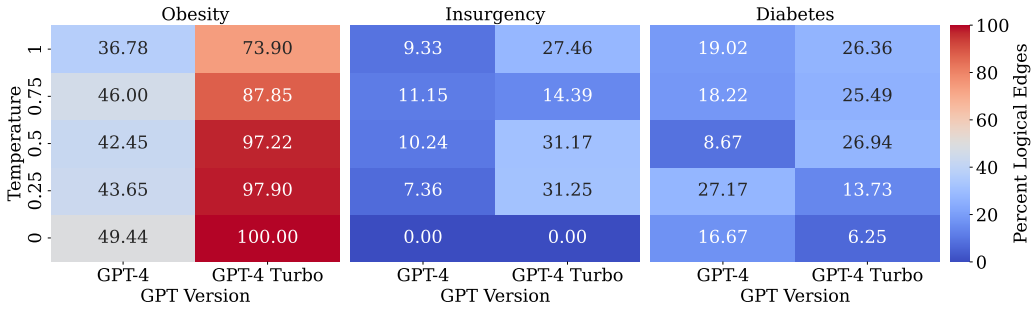

Figure 9: Mean percent of logical edges.

models, leveraging their advanced capabilities in tackling intricate tasks such as constructing knowledge graphs. We noted that the results depend on the interplay of specific LLMs (e.g., GPT-4 vs. GPT-4 Turbo), temperature setting, and case study. Best performances were achieved with a non-zero temperature setting, which indicates that some creativity contributes to resolving cognitive dissonance. In the process of resolving dissonance, the LLMs primarily added relationships among existing concepts rather than adding both new concepts and relationships. Adding more relationships did not necessarily lead to better results, which indicates that fixing dissonance is not a mere matter of saturating a model but rather a careful balancing of relevant constructs.

There are three main limitations to our work. First, we found that the LLMs rarely added new nodes and tended to only add such nodes between the dissonant concepts and existing ones in the original FCM. As a result, the LLMs may have been over-focused on adding edges, resulting in several proposed edges that were not justifiable in the application domain. This may be explored in future work by using a *scoring function* to guide the model in striking a balance between the number of new edges and nodes. The literature on Graph Edit Distance provides several suggestions for such scores [65], as it has a long history in examining how an input graph can be transformed into a given target based on operations that include adding/deleting nodes or edges [66, 67].

Second, the performance of our algorithm depends on the case study, model choice, and temperature setting. The optimal temperature varies from case to case, which means that users of our algorithm would need to optimize the process in each case. Such optimization is a routine task in applied ma-

chine learning research, where users evaluate several algorithms (e.g., deep neural networks, decision trees, random forests) and optimize them thoroughly for each application. However, it is a time consuming process and it may create a financial barrier, as users may incur a cost when sending multiple requests to LLMs. This problem is compounded in our case by the stochastic nature of the algorithm. For example, GPT-4 with temperature 0.75 on the insurgency case study would often resolve under 10% of the dissonance (Figure 10), so the LLM may have to be repeatedly queried (with the same instructions) to arrive at its optimal performance of resolving over 90% of the dissonance (Figure 11). It would thus be of particular interest to narrow down the search space and reduce output variability by identifying which characteristics of a case study tend to be associated with certain models or temperature settings.

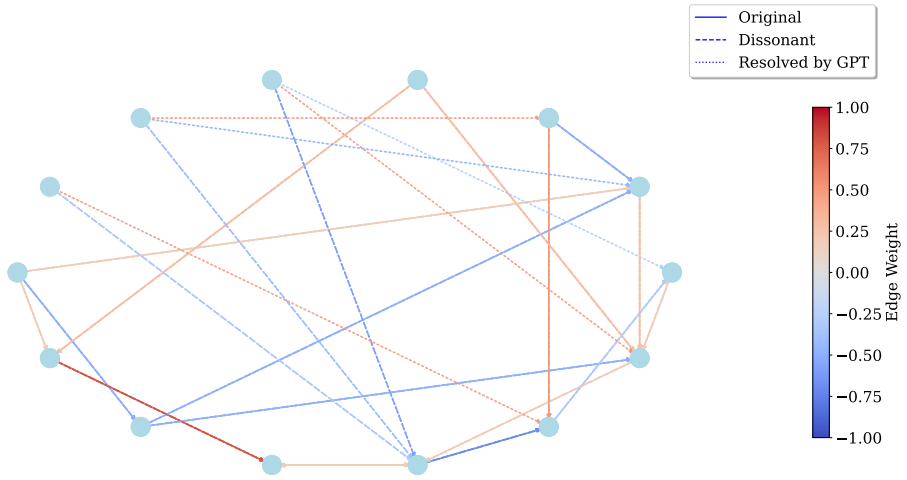

Figure 10: Original insurgency FCM, with dissonance introduced, and after implementing changes suggested by GPT-4 and a temperature value of 0.75. Under 10% of the dissonance was resolved.

Third, we observed that adding logical ideas into a sound expert system can easily trigger dissonance, and LLMs are not always able to solve this problem logically. When the LLM fails to restore consistency to an expert system, it suggests an area where the *LLM itself needs additional knowledge*. This is valuable information, as it identifies specific needs to improve an

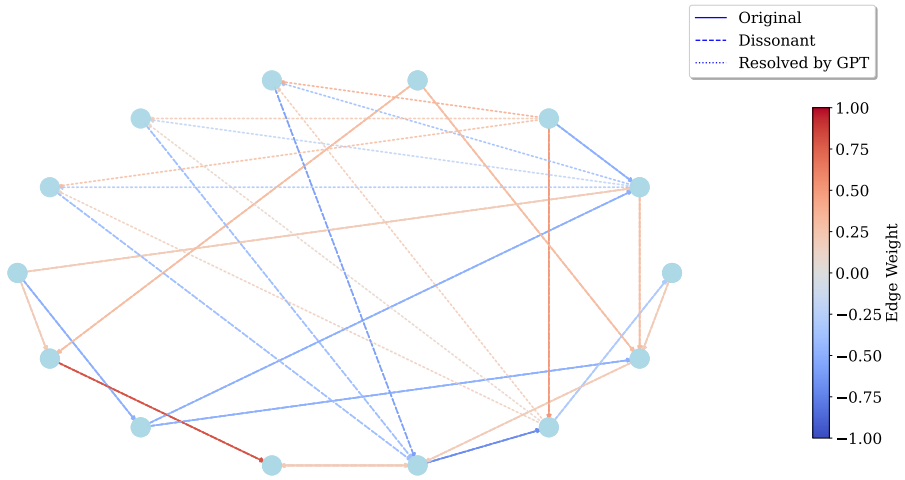

Figure 11: Original insurgency FCM, with dissonance introduced, and after implementing changes suggested by GPT-4 and a temperature value of 0.75. Over 90% of the dissonance was resolved.

LLM, which could be addressed by editing the LLM (see Section 2.4). While our study focused on using LLMs to assist with expert systems, future work can thus explore a reciprocal approach in which the early struggles of LLMs to edit an expert system would ultimately strengthen the LLM.

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Appendix A. Results Tables with Standard Error

Table A.4: Mean percent dissonance resolved ± the standard error.

Model	Case study	Temp 0	Temp 0.25	Temp 0.5	Temp 0.75	Temp 1
GPT-4	Obesity	85.45 ± 2.67	**85.50 ± 2.92**	77.38 ± 3.61	80.47 ± 4.77	76.9 ± 4.55
GPT-4 Turbo	Obesity	44.66 ± 2.61	46.81 ± 2.68	47.63 ± 2.79	49.01 ± 2.81	**56.73 ± 3.12**
GPT-4	Insurgency	0.00 ± 0.00	57.48 ± 10.96	53.46 ± 6.43	44.36 ± 5.88	**76.11 ± 6.2**
GPT-4 Turbo	Insurgency	0.00 ± 0.00	**99.21 ± 0.00**	70.76 ± 4.21	37.41 ± 4.93	65.36 ± 3.84
GPT-4	Diabetes	91.63 ± 1.72	62.03 ± 3.78	**91.67 ± 2.08**	54.14 ± 3.27	45.8 ± 2.85
GPT-4 Turbo	Diabetes	**83.27 ± 4.79**	73.00 ± 4.37	58.04 ± 3.5	60.62 ± 3.64	48.99 ± 2.96

Table A.5: The average number of new edges and nodes ± the standard error.

Model	Case study	Temp 0	Temp 0.25	Temp 0.5	Temp 0.75	Temp 1
GPT-4	Obesity	E: 4.45 ± 0.20 N: 0.00 ± 0.00	E: 4.64 ± 0.31 N: 0.00 ± 0.00	E: 4.82 ± 0.23 N: 0.00 ± 0.00	E: 6.55 ± 1.43 N: 0.00 ± 0.00	E: 4.88 ± 0.52 N: 0.08 ± 0.08
GPT-4 Turbo	Obesity	E: 3.00 ± 0.00 N: 0.00 ± 0.00	E: 3.08 ± 0.12 N: 0.00 ± 0.00	E: 2.97 ± 0.03 N: 0.00 ± 0.00	E: 3.09 ± 0.16 N: 0.00 ± 0.00	E: 2.94 ± 0.13 N: 0.00 ± 0.00
GPT-4	Insurgency	E: 0.00 ± 0.00 N: 0.00 ± 0.00	E: 5.71 ± 1.19 N: 0.00 ± 0.00	E: 6.00 ± 0.59 N: 0.00 ± 0.00	E: 6.00 ± 0.34 N: 0.00 ± 0.00	E: 7.00 ± 1.26 N: 0.00 ± 0.00
GPT-4 Turbo	Insurgency	E: 0.00 ± 0.00 N: 0.00 ± 0.00	E: 6.50 ± 0.50 N: 0.50 ± 0.50	E: 3.89 ± 0.37 N: 0.16 ± 0.09	E: 2.82 ± 0.12 N: 0.00 ± 0.00	E: 3.37 ± 0.22 N: 0.18 ± 0.11
GPT-4	Diabetes	E: 4.00 ± 0.00 N: 0.00 ± 0.00	E: 4.17 ± 0.28 N: 0.00 ± 0.00	E: 4.60 ± 0.87 N: 0.00 ± 0.00	E: 4.63 ± 0.51 N: 0.02 ± 0.01	E: 4.42 ± 0.36 N: 0.09 ± 0.04
GPT-4 Turbo	Diabetes	E: 1.33 ± 0.23 N: 0.00 ± 0.00	E: 1.37 ± 0.13 N: 0.00 ± 0.00	E: 1.53 ± 0.14 N: 0.00 ± 0.00	E: 1.45 ± 0.15 N: 0.00 ± 0.00	E: 1.59 ± 0.13 N: 0.01 ± 0.01

Table A.6: Mean percent logical edges ± the standard error.

Model	Case study	Temp 0	Temp 0.25	Temp 0.5	Temp 0.75	Temp 1
GPT-4	Obesity	**49.44 ± 4.17**	43.65 ± 4.54	42.45 ± 2.49	46.0 ± 5.58	36.78 ± 5.11
GPT-4 Turbo	Obesity	**100.00 ± 0.00**	97.90 ± 1.20	97.22 ± 1.97	87.85 ± 2.85	73.90 ± 3.76
GPT-4	Insurgency	0.00 ± 0.00	7.36 ± 2.85	10.24 ± 3.06	**11.15 ± 2.47**	9.33 ± 6.09
GPT-4 Turbo	Insurgency	0.00 ± 0.00	**31.25 ± 2.08**	31.17 ± 1.54	14.39 ± 5.04	27.46 ± 3.58
GPT-4	Diabetes	16.67 ± 0.00	**27.17 ± 1.90**	8.67 ± 3.74	18.22 ± 2.33	19.02 ± 1.97
GPT-4 Turbo	Diabetes	6.25 ± 3.45	13.73 ± 3.45	**26.94 ± 3.99**	25.49 ± 4.29	26.36 ± 3.34